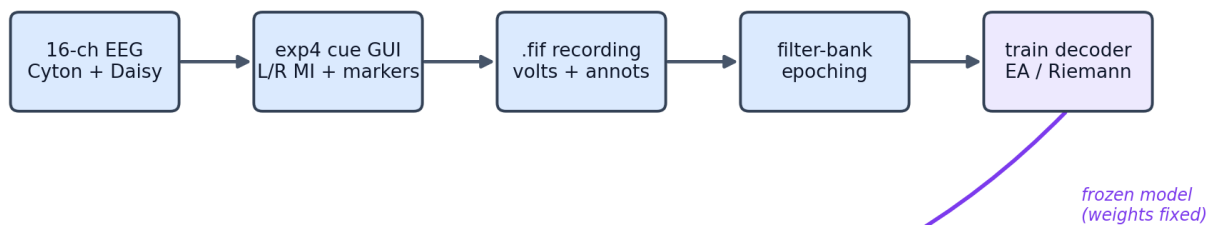


EEG Motor-Imagery BCI

End-to-end pipeline of the **main** branch — acquisition, signal processing, the two decoders (EA + FB-CSP and Riemannian tangent-space), unsupervised online adaptation, and decision smoothing.

This document describes exactly how a left/right hand motor-imagery intent travels from scalp voltages to a committed robot command. Every stage is given with its governing equations and the concrete constants used in `config.py`. Two interchangeable decoders are documented in full. All quantitative results shown come from leave-one-subject-out (LOSO) evaluation on the project's own recordings.

OFFLINE · model training



ONLINE · live decoding

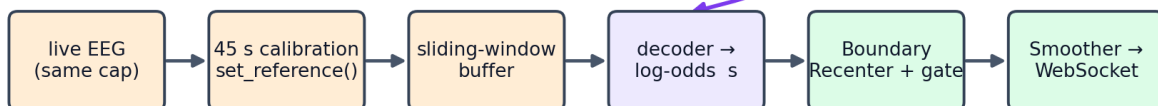


Figure 1 — System overview. Offline, cued recordings train a frozen decoder; online, the same cap streams live EEG through calibration, the frozen decoder, unsupervised recentering, and a temporal smoother that broadcasts committed decisions. Model weights never change after training.

1 Hardware and Signal Acquisition

The sensor is an **OpenBCI Cyton + Daisy** stack providing 16 EEG channels sampled at $f_s = 125 \text{ Hz}$. Data reach the host over a 2.4 GHz RFDuino radio to a USB dongle, read through the brainflow library. Fifteen channels are used for decoding; physical channel 8 is a dead electrode (DEAD) and is excluded. Electrodes follow the international 10-20 layout, with the motor strip (C3 / Cz / C4) carrying most of the sensorimotor-rhythm information that motor imagery modulates.

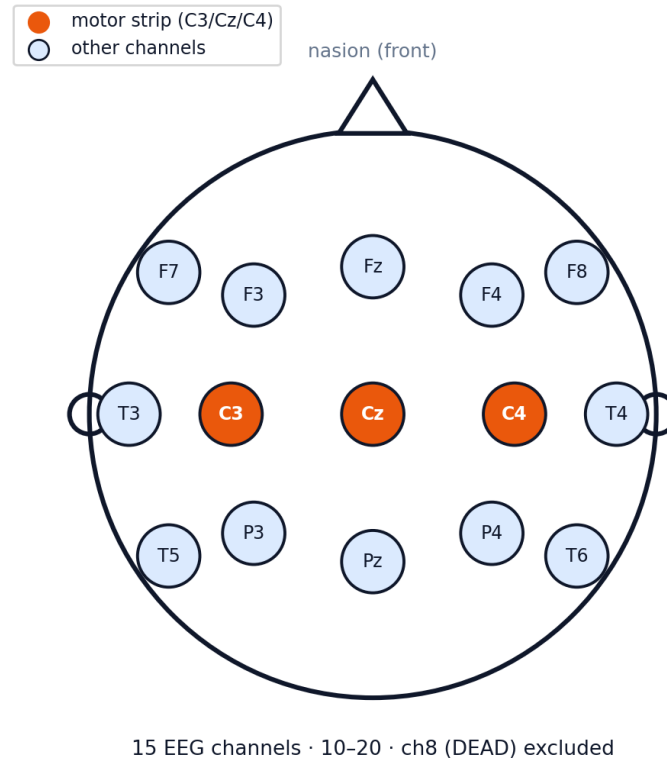


Figure 2 — Electrode montage. 15 active channels in 10-20 coordinates; the C3/Cz/C4 motor strip (orange) dominates the mu (8–13 Hz) and beta (13–30 Hz) desynchronisation during imagery.

1.1 Acquisition loop

A background thread drains the board buffer once per **stride** (0.2 s). Each wake-up calls `get_board_data()`, which returns every sample accumulated since the last call (~25 samples) and hands the chunk to a callback. The callback appends the new samples to a rolling buffer and, when enough context exists, runs one decode. Because decoding reads only the newest fixed-length window, decode latency never corrupts the signal — it only delays a decision. On weak single-board computers the FTDI latency timer must be set to 1 ms to avoid dropped samples, which otherwise inject discontinuities into the causal filters.

2 Data Collection (training paradigm)

Training data come from the **exp4** cued paradigm. The subject is shown a cue and performs kinaesthetic motor imagery of the left or right hand across four phases — *prep* → *plan* → *task* → *rest*. Every event is stamped into the stream as an integer marker encoding label, phase and trial index:

$$\text{marker} = \text{label} \times 10^6 + \text{phase} \times 10^5 + \text{trial}$$

so a marker decodes to a string such as `right_hand/task/t12`. Only the **task** phase of each hand is used as a class: the target map is `left_hand/task` → 1, `right_hand/task` → 2.

2.1 On-disk format

A recorder thread samples all devices at the board rate and, on stop, writes an MNE Raw to a `*_mi_raw.fif` file. EEG is stored in volts (the raw microvolt counts divided by 10^6), the standard-1020 montage is attached, the DEAD channel is typed as *misc*, and the integer markers are converted to timed *annotations*. This is the exact format later re-used by the online session recorder, so live sessions can be replayed through the same code path that reads the training files.

3 Preprocessing and Epoching

3.1 Filter bank

The signal is decomposed into four overlapping frequency bands spanning the mu and beta ranges where sensorimotor (de)synchronisation lives:

$$\mathcal{B} = \{ (8, 12), (11, 15), (14, 20), (20, 30) \} \text{ Hz}$$

Each band is realised as a finite-impulse-response (FIR) filter designed by the window method (`firwin`) with 2.0 Hz / 2.5 Hz transition bands. Crucially the phase is **minimum-phase**, i.e. strictly causal — the filtered sample at time n depends only on past input, which is mandatory for real-time use:

$$x_b[n] = \sum_{k=0}^{L_b-1} h_b[k] x[n-k]$$

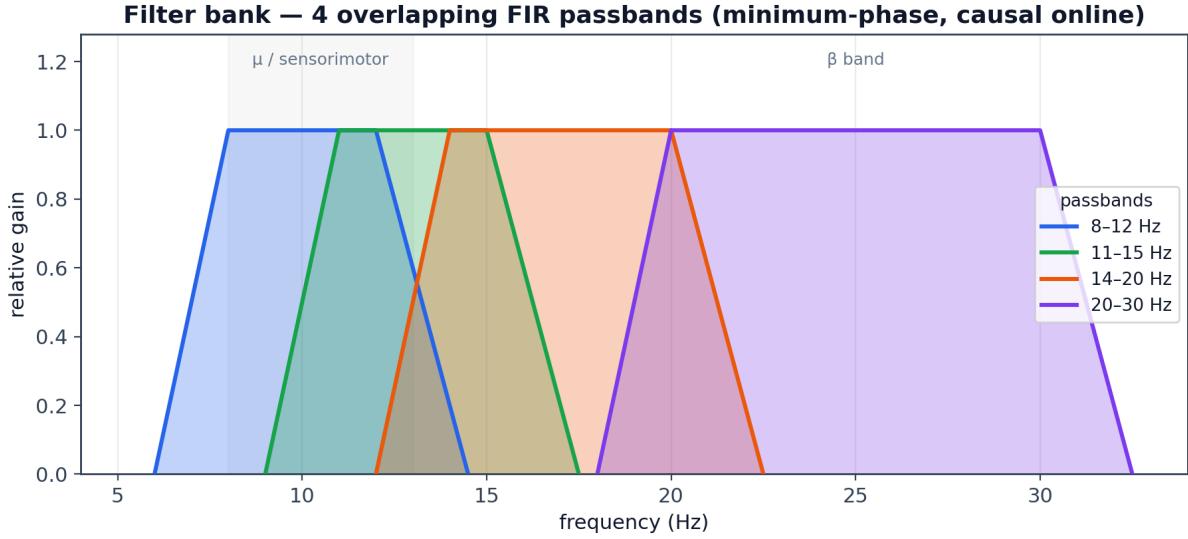


Figure 3 — The four causal FIR passbands. Overlap is intentional: neighbouring bands share spectral edges so no discriminative rhythm falls into a gap.

3.2 Epoching and artifact rejection

For each band the continuous recording is cut into fixed epochs around each task cue, from $t_{\min} = 0.5 \text{ s}$ to $t_{\max} = 2.5 \text{ s}$ post-cue — a 2.0 s window (251 samples at 125 Hz) that skips the initial cue-evoked transient. Stacking the four bands yields the per-trial tensor

$$X \in \mathbb{R}^{N \times 4 \times 15 \times 251} \quad (\text{trials} \times \text{bands} \times \text{channels} \times \text{time})$$

An epoch is dropped if its broadband (8–30 Hz) peak-to-peak amplitude on any channel exceeds 100 μV , removing blinks, motion and electrode pops before they reach the model.

4 Online Sliding-Window Buffering

Online there are no cues, so the epoch is replaced by a **rolling window** over the live stream. The buffer holds the most recent $B = 501$ samples (4.01 s): a **250-sample (2.0 s) warm-up** that lets the causal FIR settle, followed by the **251-sample (2.01 s) classify window** that matches the training epoch length. Every stride the buffer advances by 25 samples and one decode is produced, giving a steady **5 Hz** decision rate.

$$B = W + W_{\text{warm}} = 251 + 250 = 501, \quad \text{rate} = 1/\Delta t = 5 \text{ Hz}$$

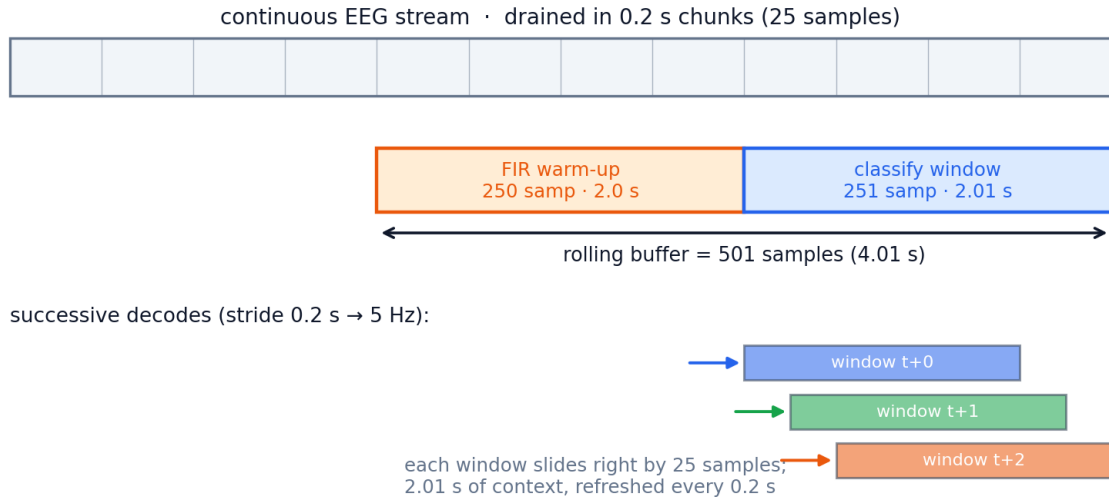


Figure 4 — Sliding-window buffering. The causal filter bank is applied to the whole 501-sample buffer; only the last 251 samples (2.01 s) become the feature window, refreshed every 0.2 s.

5 Decoder A — EA + FB-CSP (2 components)

This is the default decoder: Euclidean Alignment (EA) followed by filter-bank Common Spatial Patterns and a shrinkage LDA. It is fast (sub-millisecond per window) and transfers well across subjects.

5.1 Euclidean Alignment

Cross-subject and cross-session, the covariance structure of EEG shifts. EA removes that shift by whitening every trial with the inverse square-root of the *mean* covariance of its domain. Per band, the trial covariance and reference are

$$C_i = \frac{1}{T} X_i X_i^\top, \quad R = \frac{1}{N} \sum_{i=1}^N C_i$$

With the eigendecomposition $R = V \Lambda V^\top$ the whitener is $R^{-1/2} = V \Lambda^{-1/2} V^\top$, and each trial is aligned so that the mean covariance of the aligned set becomes the identity:

$$\tilde{X}_i = R^{-1/2} X_i \Rightarrow \frac{1}{N} \sum_i \tilde{C}_i = I$$

During training the whitener is computed **per group** (each subject/session aligned to its own identity). At deployment `set_reference()` recomputes R from the live calibration block only — an unsupervised, label-free adaptation to the current session.

5.2 CSP features and LDA

On the aligned data a Common Spatial Patterns filter is learned per band (Ledoit-Wolf regularised), keeping the **2** most discriminative components. The feature is the log-variance of each projected component; four bands $\times$ two components give an 8-dimensional vector, classified by a shrinkage LDA whose signed output is the log-odds toward class 2:

$$f_{b,k} = \log(\text{var}(w_{b,k}^\top \tilde{X})), \quad s = w^\top f + b = \log \frac{P(c_2 | f)}{P(c_1 | f)}$$

6 Decoder B — Riemannian Tangent-Space

The alternative decoder treats each trial as a point on the manifold of symmetric positive-definite (SPD) covariance matrices and classifies in its tangent space. It is often a little more robust cross-subject, at higher per-window cost, and requires `pyriemann`.

Per band a Ledoit-Wolf covariance C_i is formed. Distances use the affine-invariant Riemannian metric, whose Fréchet mean defines the recentering reference:

$$\delta_R(A, B) = \|\log(A^{-1/2} B A^{-1/2})\|_F, \quad \bar{C} = \operatorname{argmin}_M \sum_i \delta_R(M, C_i)^2$$

Each trial is recentred by the mean (the manifold analogue of EA), then mapped to the tangent space at the identity by the matrix logarithm and vectorised; a logistic regression on those tangent vectors produces the log-odds:

$$\tilde{C}_i = \bar{C}^{-1/2} C_i \bar{C}^{-1/2}, \quad v_i = \operatorname{upper}(\log \tilde{C}_i), \quad s = w^\top v + b$$

As with EA, `set_reference()` recomputes the reference — here the Riemannian mean $\bar{C}$ — from the calibration block. The two decoders expose an identical API (`fit` / `set_reference` / `predict_proba` / `analyze`) and are therefore interchangeable at runtime; everything downstream is agnostic to the choice.

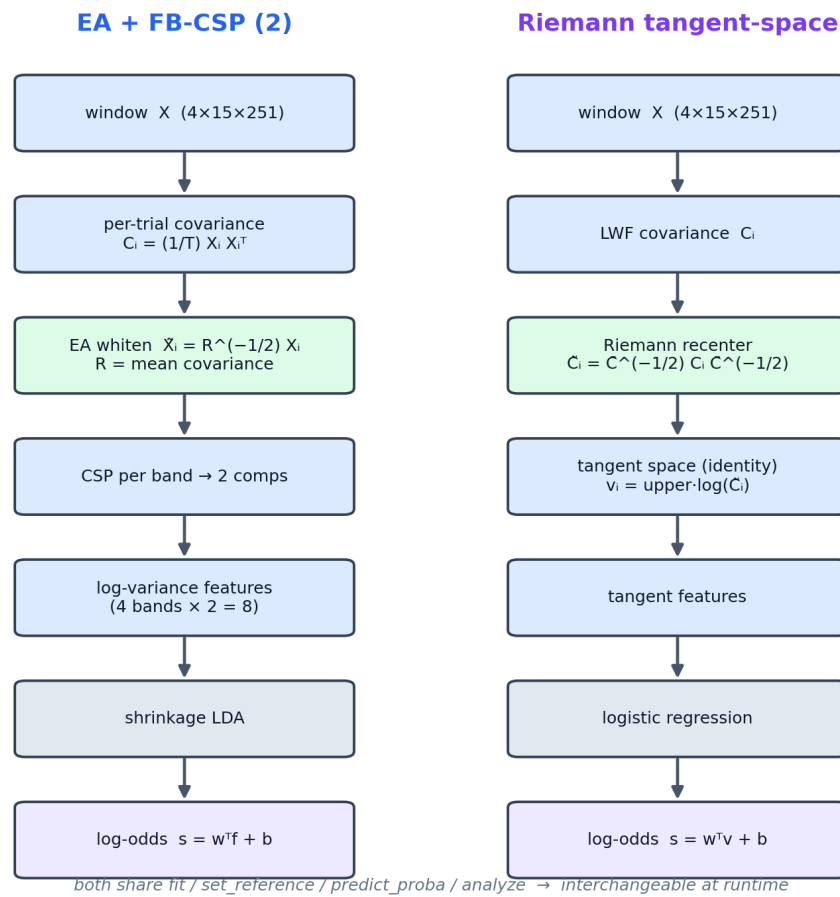


Figure 5 — The two decoders side by side. Both consume the same window and emit the same log-odds s ; they differ only in how covariance is aligned and projected to features.

6.1 Which to use

Property	EA + FB-CSP (2)	Riemann tangent-space
Alignment	Euclidean mean, $R^{(-1/2)}$	Riemannian (Fréchet) mean

Features	log-variance of 2 CSP comps/band	tangent vectors of SPD covariances
Classifier	shrinkage LDA	logistic regression
LOSO transfer (S4→S5)	94.1%	95.8%
Per-window cost	very low (matmul)	higher (cov + tangent map)
Dependency	numpy / scikit-learn	adds pyriemann
SBC deployment	yes (pure numpy runtime)	desktop only

7 Online Unsupervised Adaptation

7.1 Calibration and reference alignment

A live session opens with a **45 s calibration**: the subject performs the same imagery in the online context while the stream is sliced (every 0.5 s) into ~ 84 filter-bank windows. `set_reference()` then recomputes the alignment reference (EA mean R, or Riemannian mean C) from these windows — absorbing the train→online distribution shift *without labels* and without touching the frozen decoder weights.

7.2 Adaptive decision-boundary recentering

Alignment fixes the *covariance* but not *where the classifier's boundary sits*. Cross-subject, some subjects' log-odds distributions are shifted, so a neutral/rest state reads as a confident class — the decoder skews one way and, because the boundary is frozen while EEG drifts within a session, the skew grows until it gets stuck. LOSO makes this concrete: for the worst subject, right-hand accuracy is only 0.35 with mean confidence 0.92 on the mislabelled trials — the model is *confidently wrong*, so no fixed confidence gate can rescue it.

The fix is an unsupervised, adaptive **BoundaryRecenter**. It seeds a neutral point from the calibration median of the log-odds, decides on the recentred margin, and reads confidence as the distance of that margin from neutral:

$$c_0 = \text{median}_j s_j^{\text{cal}}, \quad z_t = s_t - c_t, \quad \hat{y}_t = c_2 \text{ if } z_t > 0, \text{ else } c_1$$

$$\text{conf}_t = \sigma(|z_t|) = \frac{1}{1 + e^{-|z_t|}}$$

The neutral **tracks slow drift** but is updated only from *rest-like* windows (small margin), so sustained real imagery never drags the boundary, and it is clamped near its seed so it cannot run away:

$$c_{t+1} = c_t + \alpha (s_t - c_t) \text{ if } |z_t| < \tau_{\text{rest}}, \quad c_{t+1} \leftarrow \text{clip}(c_{t+1}, c_0 - \Delta, c_0 + \Delta)$$

Because confidence is now measured from the recentred margin, the commit gate doubles as a **rest dead-zone**: a window commits only when it is far enough from neutral —

$$\text{commit} \Leftrightarrow \text{conf}_t \geq \theta \Leftrightarrow |z_t| \geq \text{logit}(\theta) \approx 1.735 \quad (\theta = 0.85)$$

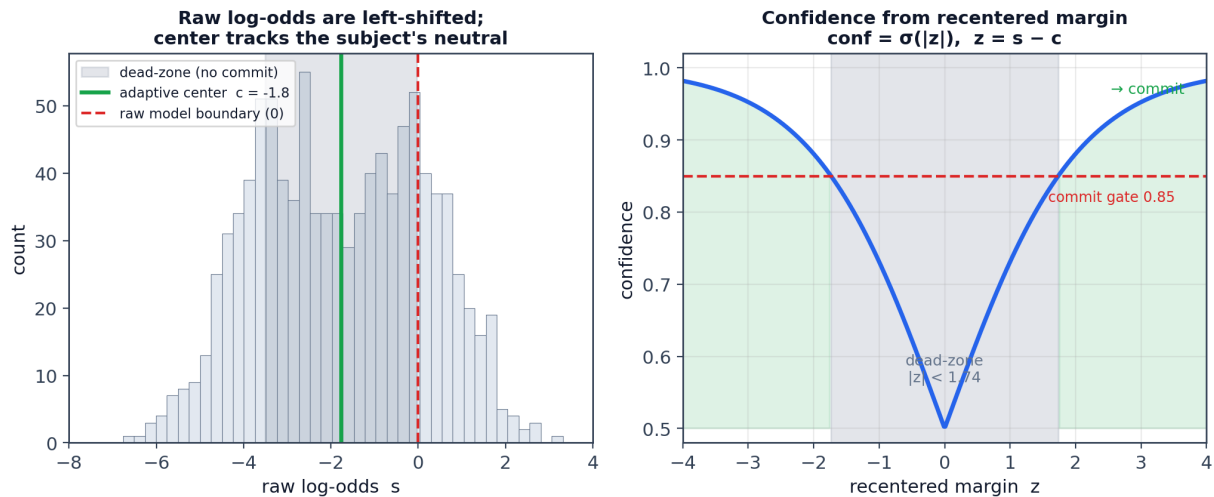


Figure 6 — Left: a left-shifted log-odds distribution; the adaptive center c sits at the subject's neutral (green), not the raw model boundary (red). Right: confidence $\sigma(|z|)$; the 0.85 gate creates a dead-zone around neutral so idle/rest never commits.

Validated by LOSO with the deployed class: the skewed subject's right-hand accuracy rises from 0.35 to 0.59 and predicted-left drops from 76% to 50%, while already-balanced subjects are untouched and overall accuracy is unchanged (0.834).

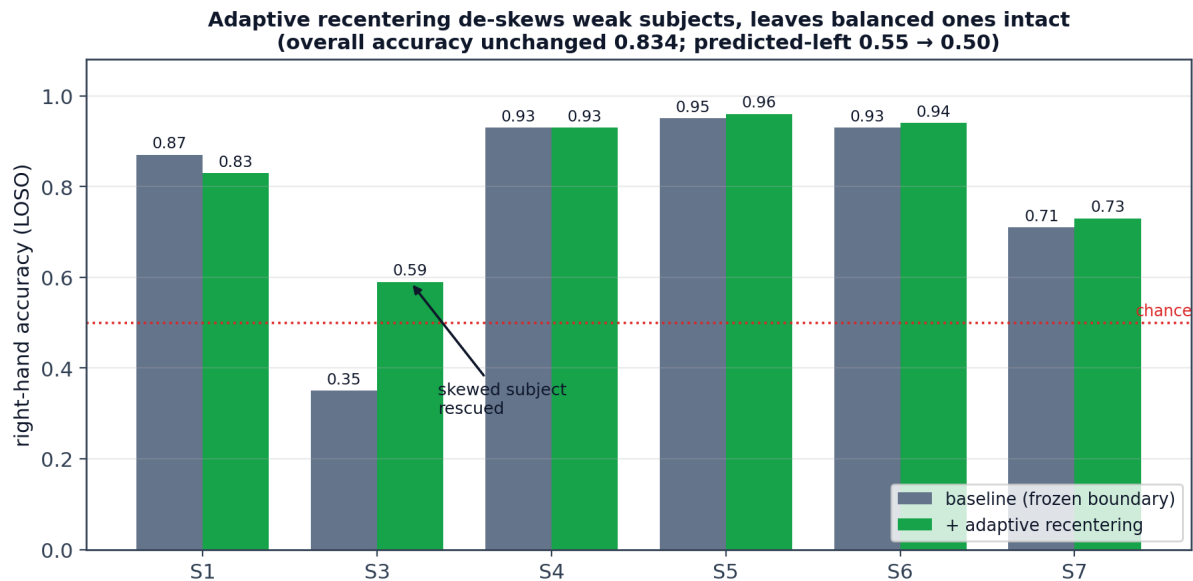


Figure 7 — Adaptive recentering rescues the skewed subject (S3) and leaves balanced subjects (S4–S6) intact; total accuracy and the class balance are corrected without retraining.

8 Temporal Smoothing and Commitment

Raw 5 Hz decisions are noisy, so a two-stage smoother converts them into stable robot commands. First a **confidence gate**: a window abstains if its recentred confidence is below $\theta = 0.85$. Surviving votes feed an **M-of-N** majority over the last $N = 5$ windows — a class becomes the consensus only with at least $M = 4$ votes, which tolerates one dissenting window. Finally a **dwell** gate: the consensus must persist for $D = 3$ consecutive windows before it is committed as the final decision and broadcast over a WebSocket to the robot/GUI.

$$v_t = \hat{y}_t \text{ if } \text{conf}_t \geq \theta, \text{ else } \emptyset \quad \text{consensus } c^* : |\{v_{t-N+1:t} = c^*\}| \geq M$$

commit $c^* \Leftrightarrow c^*$ is consensus for D consecutive windows

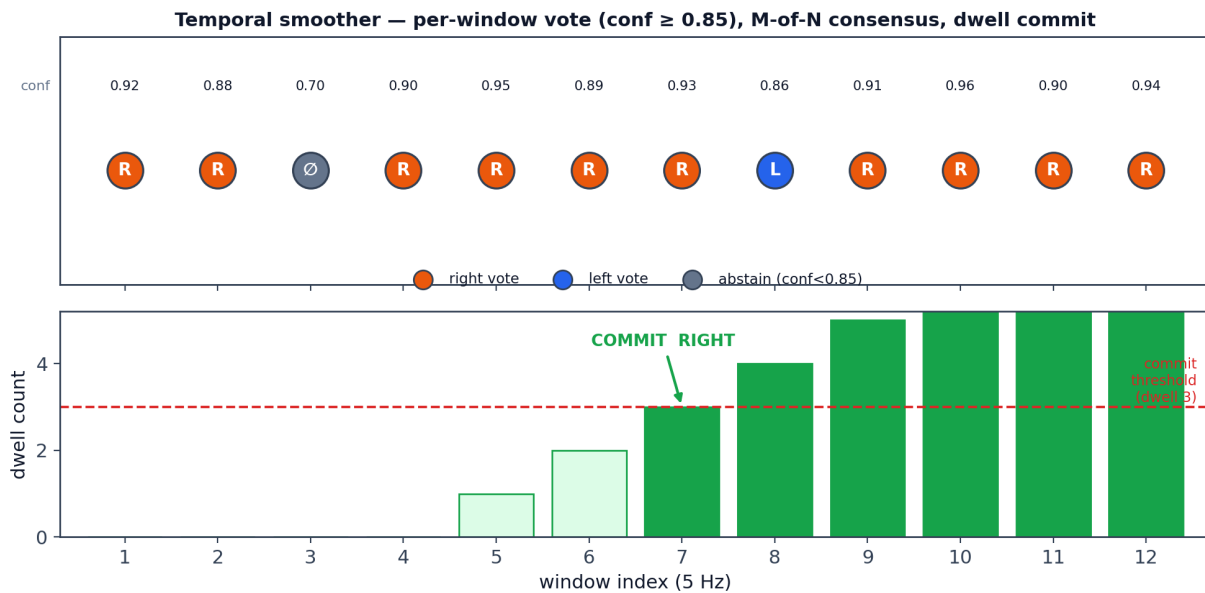


Figure 8 — Smoother in action. Per-window votes (top) with one abstain and one outlier; the dwell counter (bottom) reaches 3 and commits RIGHT. The 4-of-5 rule absorbs the single left outlier without breaking consensus.

9 Background Session Recording

In GUI live mode every session is recorded in the background for replay and analysis, without touching the real-time path (the acquisition thread only appends to in-memory lists; files are written at teardown). Two artefacts are produced:

- **online_<timestamp>_raw.fif** — the raw 16-channel EEG in the *same format as the training files*, so a session drops straight back into the pipeline for replay.
- **online_<timestamp>_decisions.csv** — a per-window decoder trace (time, log_odds, center, margin, pred, conf, committed) that lets the skew/drift behaviour be studied directly against the raw signal.

10 Configuration Reference

Every tunable lives in `config.py` as the single source of truth, imported by the desktop app and bundled into the single-board-computer binary.

Constant	Value	Meaning
STRIDE_S	0.2 s	decode/stride period → 5 Hz decision rate
FILTER_WARMUP_S	2.0 s	causal-FIR warm-up prepended to each window
EPOCH_TMIN / TMAX	0.5 / 2.5 s	training epoch bounds (2.0 s window)
FB_BANDS	8-12,11-15,14-20,20-30	four filter-bank passbands (Hz)
FB_TRANS	2.0 / 2.5 Hz	FIR transition bandwidths
CSP_COMPONENTS	2 (deployed)	CSP components kept per band
EPOCH_REJECT	100 μ V p2p	broadband artifact rejection threshold
CALIBRATION_SECONDS	45 s	live reference-alignment block length
CONF_FLOOR	0.85	commit gate / rest dead-zone on recentred conf
RECENTER_ALPHA	0.01	EMA rate of the adaptive neutral (rest windows)
RECENTER_CLAMP	2.0	max drift of neutral from its seed (log-odds)
RECENTER_REST_CONF	0.65	below this recentred conf a window updates neutral
SMOOTHER_N / M / DWELL	5 / 4 / 3	M-of-N window, majority, dwell-to-commit

Appendix Symbol Glossary

Symbol	Meaning
X_i	one trial: channels $\times$ time (per band)
C_i	spatial covariance of trial i
$R, C\text{-bar}$	Euclidean / Riemannian mean covariance (alignment reference)
s	raw log-odds toward class 2 (right) from the decoder
c_t	adaptive neutral (decision boundary) at time t
$z_t = s_t - c_t$	recentred margin used for decision and confidence
$\text{sigma}(\cdot)$	logistic sigmoid; $\text{conf} = \text{sigma}(z)$
$\text{theta}, \text{tau_rest}$	commit gate (0.85) and rest-update threshold (0.65)
N, M, D	smoother window, majority, dwell-to-commit (5, 4, 3)

Generated from the main branch of `eegthingy`. Figures 7 and the accuracy figures are measured by leave-one-subject-out evaluation on the project's own exp4 recordings.